

Dingning Cao

📍 Cambridge, MA | ✉ dcao2028@mit.edu | 📞 571-519-4359 | 📁 Portfolio | 🌐 LinkedIn

Education

Massachusetts Institute of Technology

Sept 2024 – May 2028

BS in Computer Science + Arts and Design (GPA 4.7)

- Designer of MIT 2028 Brass Rat; MITIT, Reality Hack, Infinite Magazine Art Director, HackMIT Marketing
- **Relevant Coursework:** Computer Vision, Computer Graphics, Algorithms, Linear Algebra, Machine Learning

Milton Academy

Sept 2020 – June 2024

Skills

Programming: Python, C# (Unity), JavaScript, HTML/CSS, React, Java, PyTorch, scikit-learn, ComfyUI, C++

Design & Modeling: Figma (UI/UX), Adobe Suite, Procreate, Blender (3D Modeling), Maya, Rhino

Research | Cambridge, MA

MIT Bill Freeman Group

May 2026 - Present

- 3D and video generation and reconstruction of Dunhuang cave painting dance.

MIT HCIE Group Researcher (with Jiaji Li & Stefanie Mueller)

May 2025 - Present

- *Y-zipper: 3D Printing Flexible-Rigid Transition Mechanism for Rapid Assembly* (CHI 2026 Accepted)
- *Xspine: Integrating Motion Sensing Capability into Dynamic Structures Using Multi-material FDM 3D Printing* (UIST 2025 Posters Accepted, CHI 2026 Accepted)
- *ShiftLens: Mechanically Actuated Optical Surfaces for Switchable Appearances on 3D Objects* (UIST Accepted)
- *MiXR: Harvesting and Recomposing Geometry from Real-World Objects for In-Situ 3D Design* (Under Review)

MIT CSAIL Researcher (with Antonio Torralba & Yael Vinker)

Feb 2025 - Sep 2025

- *Doodle Agent: Exploring Freeform Visual Generation with Multimodal LLMs* (ICCV AI4VA Accepted)
- Developed full-stack platform for [SketchAgent](#) [🔗](#) (Flask, React, SVG), extending the original work.

Experience

Machine Learning Engineer - Apple

Summer 2026

- Built an agent for stylized slide generation, transferring arbitrary Keynote templates into style-consistent decks.
- Curated and preprocessed model training data for a layered generation model

AI Researcher - Lica World

2026

- Fine-tuned models for layout generation in poster and presentation design.
- *Evaluating Design Video Generation: Metrics for Compositional Fidelity* (ICML Workshop Accepted)

AI Technical Artist - miHoYo (Genshin Impact)

2025-2026

- Trained miHoYo-specific LoRAs and built generation pipelines in ComfyUI.
- Created an independent agent-powered tool supporting upstream design for *Genshin Impact*.

Projects

Game Development Projects

2023-2025

- [Threadspace](#) [🔗](#) (Godot, GMTK 2025) Designed mechanics, art assets, and 8 levels with OOP components; coordinated playtests. ★ Ranked #48/9600 (Top 0.5%), featured at Boston Indie Game Festival.
- [Pizzapocalypse!](#) [🔗](#) (Unity, iOS) Implemented tilemap levels, NPC AI, UI systems, and progression mechanics. ★ Published on iOS (4.8★ rating); won NHSGA Mass Appeal Prize.
- [Tea Time in Five](#) [🔗](#) (Unity, Brackeys Jam 2025, 3D) Modeled Blender art assets and coded movement systems; managed team and playtesting. ★ Ranked 236/2299.

Creative Web Applications

2024-2025

- **Dreamscape:** Full-stack journaling platform (React, Node.js, MongoDB) with media storage and collaborative editing. ★ Placed 5th in MIT WebLab.
- **Ablecademy:** iOS+iPad app built in Swift to assist children with Down syndrome; conducted user testing.

Awards

- MIT 2026 Outstanding UROP Student Award
- MIT CSAIL Agentic AI Hakathon Winner
- CHI 2026 Interactive Demo Jury Honorable Mention
- USA Computing Olympiad Gold Division — Jan 2024
- VEX Robotics Regional Skills Champion & Design Award (World #12) — Feb 2024